

This chapter presents the key code modules of the AgriTrace platform. The codebase follows a modular architecture with clear separation of concerns.

## 8.1 Firebase Configuration

```
1 import { initializeApp, getApps, getApp } from 'firebase/app';
2 import { getAuth } from 'firebase/auth';
3 import { getFirestore } from 'firebase/firestore';
4 import { getStorage } from 'firebase/storage';
5
6 const firebaseConfig = {
7   apiKey: process.env.NEXT_PUBLIC_FIREBASE_API_KEY,
8   authDomain: process.env.NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN,
9   projectId: process.env.NEXT_PUBLIC_FIREBASE_PROJECT_ID,
10  storageBucket: process.env.NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET,
11  messagingSenderId: process.env.
12    NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID,
13  appId: process.env.NEXT_PUBLIC_FIREBASE_APP_ID,
14 };
15
16 const app = !getApps().length ? initializeApp(firebaseConfig) :
17   getApp();
18 export const auth = getAuth(app);
19 export const db = getFirestore(app);
20 export const storage = getStorage(app);
```

Listing 8.1: Firebase Initialization (firebase.ts)

## 8.2 Firebase Service Layer

```
1 // Create a new waste listing
2 export async function createListing(
3   data: Omit<Listing, 'id' | 'createdAt' | 'updatedAt' | 'status'
4   '>
5 ): Promise<string> {
6   const docRef = await addDoc(collection(db, 'listings'), {
7     ...data,
8     status: ListingStatus.OPEN,
9     createdAt: serverTimestamp(),
10    updatedAt: serverTimestamp(),
11  });
12  return docRef.id;
13 }
```